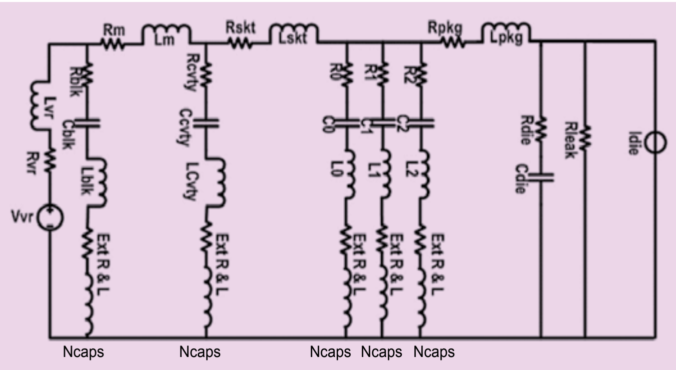

Fig. 3: Equivalent circuit model of the power delivery network

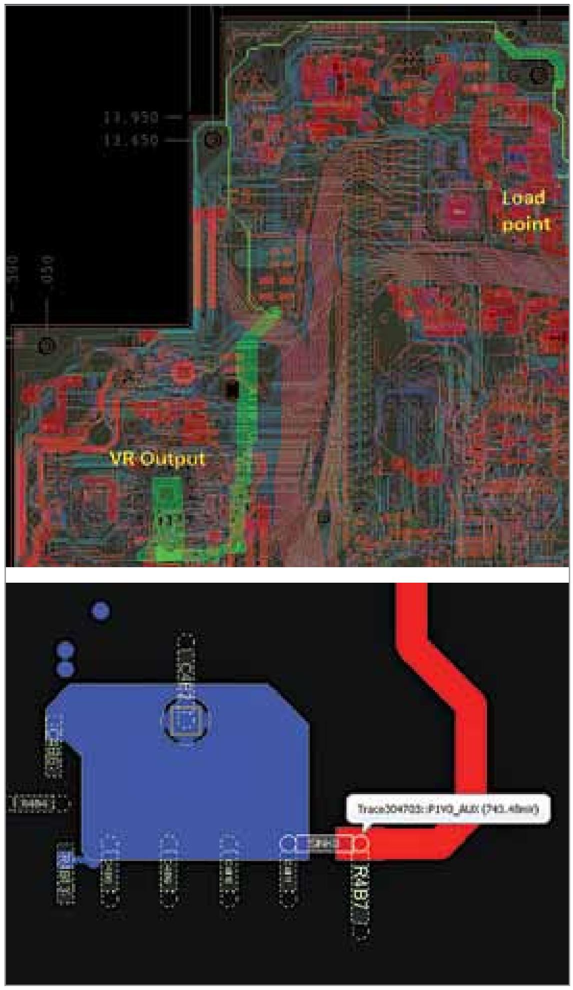

Fig. 4: PDN failure case with large voltage drop at load input

CPU is modelled as R_P. This simple PDN model can be used to understand the importance of power integrity at preliminary stages of the design.

With increasing CPU current, which may be caused by increase in CPU clock frequency to execute a specific workload, voltage V_1 at CPU input reduces due to higher voltage drop along the path or resistor R_P. Every electrical part, such as CPU, will have its maximum load demand I_{max} and minimum operating voltage V_{min} required for stable operation defined in the part datasheet. Path resistance R_P that ensures $V_1 > V_{min}$ under all loading conditions is termed as target resistance (R_{Target}) of the design.

To model the design parasitic more accurately, series inductance of the layout is considered as well. With inductance involved, it's no longer just resistance but impedance Z_P of the layout that needs to be modelled. Layout Impedance Z_P that ensures $V_1 > V_{min}$ under all loading conditions within system frequency range of operation is termed as target impedance (Z_{Target}).

The main aim of power integrity analysis is to ensure parasitic impedance Z_P of a system is less than Z_{Target} within system frequency range of operation. For any system, Z_{Target} will be determined based on its maximum load demand and maximum peak-to-peak noise that can be tolerated at input for reliable operation.

Fig. 2 shows a power delivery network of a CPU on a server or desktop motherboard system. It consists of a VR module, which is the main power source for CPU, followed by bulk capacitors at the output of the VR for limiting switching noise or ripple. There are motherboard capacitors placed close to the CPU, which are mostly multilayer ceramic capacitors (MLCC) that limit voltage falls during sudden inrush current demand by CPU.

The CPU can be connected to the motherboard via socket mechanism or soldered down directly during assembly process. The printed circuit board that mounts the silicon die or CPU chip is called a package. On the package, there are land side caps (LSC) that are placed in the pin field region at the bottom side and die side caps (DSC) that are placed right next to the silicon die.

Fig. 3 shows an equivalent circuit model of the power delivery network, including realistic circuit models of each component, such as capacitors. Power integrity analysis helps us understand the impact of each component and its parasitic value on the final performance of the CPU and helps in designing an optimised system for reliable operation.

Power integrity analysis methodology

With understanding of the importance, lets discuss the process to be followed for complete PI analysis of any given system. We start with DC performance check of the system followed with AC or frequency domain analysis (FDA). At the end, we must perform the transient analysis that provides final go or no-go direction for the design and is the ultimate test for system reliability.

DC performance analysis. DC performance is #1 priority for any